

MODULE 04

Data Collection and Preprocessing

AI with Data Science - Professional Learning Series

ARSHITH LMS | Professional Edition

Data Collection and Preprocessing

Data Sources

Structured: CSV, Excel, SQL databases
Unstructured: Text, images, audio, video
Semi-structured: JSON, XML, HTML

Collection Methods:

Web scraping (BeautifulSoup, Scrapy)
APIs (REST, GraphQL)
Databases (SQL queries)
Surveys and forms
Sensors and IoT devices

Loading Data

```
import pandas as pd

# CSV
df = pd.read_csv('data.csv')

# Excel
df = pd.read_excel('data.xlsx', sheet_name='Sheet1')

# SQL
import sqlite3
conn = sqlite3.connect('database.db')
df = pd.read_sql('SELECT * FROM table', conn)

# JSON
df = pd.read_json('data.json')
```

Handling Missing Values

```
import pandas as pd

# Detect missing values
df.isnull().sum()
df.isnull().mean() * 100 # percentage missing

# Drop missing
df.dropna() # drop rows
df.dropna(axis=1) # drop columns
df.dropna(thresh=5) # keep if 5+ non-null

# Fill missing
df.fillna(0) # fill with value
df['age'].fillna(df['age'].mean()) # fill with mean
df['city'].fillna(method='ffill') # forward fill
```

Data Cleaning

```
# Remove duplicates
df.drop_duplicates()

# Fix data types
df['date'] = pd.to_datetime(df['date'])
df['age'] = df['age'].astype(int)

# String cleaning
df['name'] = df['name'].str.strip().str.lower()
df['name'] = df['name'].str.replace('[^a-z ]', '', regex=True)

# Outlier removal (IQR method)
Q1 = df['price'].quantile(0.25)
Q3 = df['price'].quantile(0.75)
IQR = Q3 - Q1
df = df[(df['price'] >= Q1-1.5*IQR) & (df['price'] <= Q3+1.5*IQR)]
```

Feature Engineering

```
# Encoding categorical variables
from sklearn.preprocessing import LabelEncoder, OneHotEncoder
le = LabelEncoder()
df['gender_enc'] = le.fit_transform(df['gender'])

# One-hot encoding
df = pd.get_dummies(df, columns=['city'])

# Scaling/Normalisation
from sklearn.preprocessing import StandardScaler, MinMaxScaler
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)

# Create new features
df['age_squared'] = df['age'] ** 2
df['price_per_sqft'] = df['price'] / df['area']
```